Vishnuram Arumugam

Curriculum Vitae

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Research Interests

Group theory, graph theory, probability and combinatorics, with particular interests in permutation groups and symmetries of graphs.

Education

2022–2026 **PhD (Mathematics)**
The University of Western Australia
Thesis Title: Groups of Lie Type Acting on Generalised Quadrangles
Description: Studying the existence of generalised quadrangles that admit a point- and line-primitive automorphism group with socle isomorphic to a Suzuki or a Ree group.
Supervisors: Michael Giudici and John Bamberg.

2016–2021 **Bachelor of Engineering Honours (Mechatronic Engineering)**
Bachelor of Science (Computer Science)
Curtin University
Thesis Title: A Geometric Framework for Distributed Frequency Models
Description: Establishing geometric characterisations of control theoretic notions for state-space systems with a fractional derivative operator.
Supervisors: Lorenzo Ntogramatzidis and Fabrizio Padula.

Publications and Preprints

[1] Arumugam V. (2026). "Large Ree groups acting point and line-primitively on finite generalised quadrangles".
arXiv: https://arxiv.org/abs/2610.39690

[2] Arumugam V., Praeger C. E., Rademacher D. (2026). "Structure of Cayley Codes".
arXiv: https://arxiv.org/abs/2606.28611

[3] Arumugam V., Bamberg J., & Giudici M. (2025). "Low Rank Groups of Lie Type Acting Point- and Line-Primitively on Finite Generalised Quadrangles". In *Electronic Journal of Combinatorics*, 32(1), Article P143. https://doi.org/10.37236/12993.

[4] Arumugam V., Dietrich H., & Glasby S. P. (2024). "Derangements in wreath products of permutation groups". In *Journal of Algebraic Combinatorics*, 59(1), 1-22. https://doi.org/10.1007/s10801-023-01255-1.

[5] Arumugam V., Ferrante A., Ntogramatzidis L., & Padula F. (2024). "A geometric framework for distributed frequency models". In *Communications in Nonlinear Science and Numerical Simulation*, 136. https://doi.org/10.1016/j.cnsns.2024.108088.

[6] Arumugam V., Ntogramatzidis L., & Ferrante A. "On the structure of the solutions of continuous-time algebraic Riccati equations with closed-loop eigenvalues on the imaginary axis". In *Proceedings of the 21st International Federation of Automatic Control (IFAC) World Congress*, Berlin, Germany, July 11-17, 2020.

Articles in Preparation

[1] **Title:** Multicirculants and Generalised Quadrangles
Authors: Vishnuram Arumugam, Boštjan Frelih and Klavdija Kutnar
Description: Studying multicirculant structures on the incidence and point graphs of finite generalized quadrangles.

[2] **Title:** Point-primitive Generalised Quadrangles
Authors: Vishnuram Arumugam, John Bamberg and Tao Feng
Description: Analysing point-primitive generalised quadrangles of product action type.

[3] **Title:** Cayley Numbers
Authors: Vishnuram Arumugam and Edward Dobson
Description: Classifying cyclic numbers that are Cayley numbers.

[4] **Title:** Point-regular Generalised Quadrangles
Authors: Vishnuram Arumugam and Zhaochen Ding
Description: Studying point-regular generalised quadrangles of order (s, s) where s is odd.

Research Visits and Collaborations

June 2026 — **Collaboration with Tao Feng**
Zhejiang University, China
Host: Professor Tao Feng, Zhejiang University
Topic: Point-primitive Generalised Quadrangles.

August 2026 — **Collaboration at UP FAMNIT**
University of Primorska, Slovenia
Host: The Faculty of Mathematics, Natural Sciences and Information Technologies
Topic: Algebraic Graph Theory.

Awards and Funding

2022–2025 — **Australian Research Training Program**
Australian Government
Funding support for students undertaking higher degree by research

2023 — **AMSI Summer School 2023 Travel Grant**
Australian Mathematical Sciences Institute
Funding for attending the AMSI Summer School 2023

2022 — **AMSI Winter School 2022 Travel Grant**
Australian Mathematical Sciences Institute
Funding for attending the AMSI Winter School 2022

Teaching Experience

2018–2026 **Tutor in Mathematics and Engineering**
Curtin University
Units: Calculus 2, Linear Algebra 2, Complex Analysis, Functional Analysis, Dynamic Modelling & Control, Advanced Mathematics 2, Mathematics & Probability Theory and Network Optimisation.
Responsibilities: Tutorials, workshops, labs, some lectures and marking.

2023–2026 **Tutor in Mathematics and Computer Science**
The University of Western Australia
Units: Discrete Structures, Algebra, Multivariable Calculus, Mathematical Theory & Methods and Mathematics Fundamentals.
Responsibilities: Tutorials, workshops, labs, some lectures and marking.

Selected Talks

May 2025 **Low Rank Groups of Lie Type Acting on Generalised Quadrangles Parts I & II**
Groups and Combinatorics Seminars
Centre for the Mathematics of Symmetry and Computation
The University of Western Australia

Sep 2023 **An Introduction to Graph Limit Theory**
Mathematics and Physics Research Talks
The University of Western Australia

Aug 2023 **History and Various Constructions of the Family of Suzuki Groups**
Groups and Combinatorics Seminars
Centre for the Mathematics of Symmetry and Computation
The University of Western Australia

Technical Skills

Programming LaTeX, Mathematica, GAP, Python, C, Magma, MATLAB, Git.

Referees

Professor Michael Giudici (PhD Supervisor)
The University of Western Australia
michael.giudici@uwa.edu.au

Associate Professor John Bamberg (PhD Co-supervisior)
The University of Western Australia
john.bamberg@uwa.edu.au

Dr Greg Gamble
The University of Western Australia
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